Speech-to-Action: A Comparative Study of Classical and Deep Learning Approaches for Spoken Intent Classification

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***Abstract*—Spoken intent classification is the task of mapping a short voice command directly to the action it asks for. In this work, the extent to which different speech representations can solve this task is studied, starting from classical short-time analysis and ending with self-supervised embeddings. The Fluent Speech Commands corpus is used, which has 30,043 utterances from 97 speakers and 31 intents. A classical feature-extraction pipeline (short-time energy, zero-crossing rate, autocorrelation pitch, linear prediction and mel-frequency cepstral coefficients, or MFCC) is first built and checked on individual utterances. These features are then summarised over each utterance and classified with a support vector machine (SVM) and a multi-layer perceptron. Dynamic time warping and per-class Gaussian hidden Markov models are used as classical sequence baselines, one-dimensional convolutional networks are trained from scratch on MFCC and log-mel sequences, and finally a frozen wav2vec 2.0 encoder is evaluated with pooled and frame-sequence classifiers. All models are compared using five-fold speaker-grouped cross-validation, both on the 31-class task and on reduced-class tasks for the action, object and location slots. Utterance statistics reach $24.0 \pm 1.0\%$ on the intent and stay close to the majority-class baseline on the slot tasks. A convolutional network on log-mel sequences reaches $79.4 \pm 6.8\%$. An ablation shows that applying the cepstral DCT to the same input lowers this to 63.5%, because per-channel standardisation inflates the low-variance coefficients. The same network on frozen wav2vec 2.0 frames reaches $93.1 \pm 3.1\%$ on the intent and 95.1–97.5% on the slots. Overall, keeping the temporal order of the utterance matters more than any change of static feature or any reduction of the label set, and the remaining errors of the best model are mostly between commands of opposite polarity.**

***Index Terms*—spoken language understanding, intent classification, linear prediction, MFCC, dynamic time warping, hidden Markov model, convolutional neural network, wav2vec 2.0**

I. Introduction

Voice assistants in the home are given short commands such as *"turn the kitchen lights off"* or *"make the music louder"*. To act on such a command, the device does not need a transcript of what was said; it needs the action that the command refers to. Spoken language understanding (SLU) systems have traditionally done this in two steps, with a speech recogniser followed by a text classifier. End-to-end SLU models, on the other hand, map the audio signal to the intent directly [1], [2]. The Fluent Speech Commands (FSC) corpus [2] was released to support this direct approach, and most of the results reported on it today come from models that are pre-trained on large speech corpora [2]–[4].

Since accuracies on FSC are now close to saturation, a different question has received less attention: how much of the task is solved by each layer of speech representation, from the short-time measurements of classical speech analysis to the contextual embeddings of a self-supervised encoder? To answer this, the representation has to be changed while the classifier and the evaluation are kept the same wherever possible. Hence, this study is organised into four tiers. The first tier uses hand-crafted features summarised over the utterance. The second applies classical sequence models (dynamic time warping and hidden Markov models) to MFCC sequences. The third trains convolutional networks from scratch on MFCC and log-mel sequences, and the fourth places the same convolutional network on top of a frozen wav2vec 2.0 encoder [5].

The main contributions of this work are as follows.

- A classical feature-extraction pipeline (energy, zero-crossing rate, autocorrelation pitch, short-time Fourier analysis and linear prediction) is implemented and checked on a single utterance, and the accuracy that prosodic statistics alone can give on FSC is measured.
- Four representation tiers are compared under controlled conditions. This includes classical sequence baselines and an ablation that separates the decorrelating and truncating effects of the cepstral DCT while keeping the convolutional network the same.
- It is shown that the official FSC partitions have too few speakers for a reliable single-split evaluation, and five-fold speaker-grouped cross-validation with fold-wise paired comparisons is used instead.
- Reduced-class tasks based on the slot structure of the labels (action, object and location) are evaluated alongside the 31-class task. The errors of the best model are also analysed in terms of how the commands are phrased and how speakers differ.

The rest of the paper is organised as follows. Section II reviews related work and Section III states the research gaps and objectives. Section IV describes the dataset and Section V the methodology. The results are discussed in Section VI, and Section VII concludes the paper.

II. Related Work

A. Short-time analysis and classical features

Rabiner and Sambur [6] combined short-time energy and zero-crossing rate to find the start and end points of isolated utterances. Atal and Rabiner [7] then treated the voiced/unvoiced/silence decision as a statistical pattern classification problem over several such measurements. Rabiner [8] analysed autocorrelation pitch detection, including limiting the lag search range to avoid octave errors; the YIN estimator [9] later reduced these errors further using a cumulative mean-normalised difference function. Allen and Rabiner [10] formalised short-time Fourier analysis and synthesis. Makhoul [11] gave the standard account of linear prediction, where the autocorrelation normal equations are solved using the Levinson–Durbin recursion. Davis and Mermelstein [12] introduced mel-frequency cepstral coefficients and showed that they perform better than parameters based on linear prediction for word recognition. Furui [13] showed the value of time-derivative cepstral features for speaker-independent recognition.

B. Classical sequence modelling

Sakoe and Chiba [14] introduced dynamic time warping (DTW) as a way of aligning utterances of different lengths with stored templates. Rabiner's tutorial [15] established the hidden Markov model (HMM) as the standard statistical framework for variable-length speech, with Gaussian emission densities over cepstral frames. Both methods were developed for isolated-word recognition with small vocabularies, which is quite similar in structure to command classification.

C. Neural acoustic models and the choice of input feature

The use of deep networks for acoustic modelling [16] changed the role of feature extraction. Mohamed *et al.* [17] found that deep belief networks work better on log mel filter-bank energies than on MFCC. They explained this by the information lost in the cepstral transform and by the network's ability to model correlated inputs. Abdel-Hamid *et al.* [18] applied convolution along the frequency and time axes of filter-bank inputs, and Sainath and Parada [19] showed that small convolutional networks work well for keyword spotting with a limited computational budget. The Speech Commands corpus [20] provided a standard benchmark for such short-command recognisers, and SpecAugment [21] introduced time and frequency masking as a way of augmenting the input features. Studies comparing feature families in other speech classification tasks [22], [23] also report that a single hand-crafted feature family reaches a plateau, and that cepstral features or a fusion of features recover much of the gap.

TABLE I
Published intent accuracy on FSC (official test split)

Work	Model	Acc. (%)
SUPERB [4]	Log mel filter bank, frozen	9.10
SUPERB [4]	wav2vec 2.0 Base, frozen	92.35
SUPERB [4]	HuBERT Base, frozen	98.34
Lugosch *et al.* [2]	End-to-end, no pre-training	96.6
Lugosch *et al.* [2]	ASR pre-trained, frozen	98.8
Radfar *et al.* [3]	End-to-end transformer	99.6

D. End-to-end SLU and self-supervised representations

Serdyuk *et al.* [1] showed that intent can be predicted directly from audio, without an intermediate transcript. Lugosch *et al.* [2] released FSC along with a model whose lower layers are pre-trained to recognise phonemes and words. This model reaches 98.8% accuracy with the pre-trained layers frozen and 96.6% when trained from random initialisation. Radfar *et al.* [3] reported 99.6% intent accuracy with an end-to-end transformer. Self-supervised encoders learn speech representations from unlabelled audio [5], [24], [25]; for example, wav2vec 2.0 [5] masks latent speech features and solves a contrastive task over quantised targets. The SUPERB benchmark [4] evaluates frozen encoders with small downstream heads, and it reports FSC intent accuracies of 9.10% for log mel filter-bank input, 92.35% for wav2vec 2.0 Base and 98.34% for HuBERT Base. Layer-wise probing [26] further shows that phonetic and word-level information is not spread evenly across the layers of wav2vec 2.0. Table I lists the published FSC results that are used for reference in Section VI.

III. Research Gaps and Objectives

From the literature reviewed above, the following gaps were identified.

1) Work on FSC is dominated by pre-trained end-to-end models [2]–[4]; the accuracy that classical, interpretable speech features can achieve on this corpus is rarely reported.
2) Cepstral features are known to outperform prosodic or formant features for word recognition [12], but this gap has not been measured with a fixed classifier on an intent task.
3) Classical sequence models (DTW, HMM) [14], [15] and neural sequence models [18], [19] are rarely compared on the same non-pre-trained input features.
4) The effect of the cepstral (DCT) step on a convolutional classifier has been studied for acoustic modelling [17], but not for utterance-level intent classification.
5) Published FSC results use the official test split of ten speakers, and how sensitive these results are to the choice of speakers is not reported.

Based on these gaps, the objectives of this study are:

1) to implement and verify a classical feature-extraction pipeline based on short-time analysis, and to measure the intent accuracy of prosodic and cepstral utterance statistics with SVM and MLP classifiers;

TABLE II
OFFICIAL FSC PARTITION

Subset	Utterances	Speakers
Train	23,132	77
Validation	3,118	10
Test	3,793	10
Total	30,043	97

2) to benchmark DTW and Gaussian HMM classifiers on MFCC sequences;
3) to train convolutional sequence models from scratch on MFCC and log-mel inputs, including a multi-head variant that follows the slot structure of the labels;
4) to evaluate a frozen wav2vec 2.0 representation with pooled and sequence-level downstream classifiers;
5) to repeat the evaluation on reduced-class slot tasks with 4, 6 and 14 classes; and
6) to compare all models using speaker-grouped cross-validation and to analyse the remaining errors.

IV. DATASET

The Fluent Speech Commands corpus [2] is used in this study. It contains 30,043 English utterances (19.0 h) recorded at 16 kHz by 97 speakers. Each utterance is labelled with three slots: *action* (6 values), *object* (14) and *location* (4). The combinations of these slots that occur in the data form 31 intents, and these intents are expressed through 248 different phrasings. The official partition is speaker-independent (Table II), and it was confirmed that no speaker appears in more than one subset before any experiment was run. The corpus is released under a non-commercial licence for academic research.

The intents are not evenly represented (Fig. 1). The two volume commands have 2,296 and 2,099 training utterances, while the least frequent intents have only 342–401. Hence, a classifier that always predicts the majority class already gets 10.4% on the test subset. The utterances are 2.31 s long on average, and 95% of them are shorter than 3.41 s. Two properties of the corpus are important for the analysis that follows. First, several intents differ only in the polarity of the action, and this polarity is often carried by a single short word (*"turn the lamp on"* versus *"turn the lamp off"*, *"volume up"* versus *"volume down"*). Second, the official validation and test subsets contain only ten speakers each.

V. METHODOLOGY

Fig. 2 shows the overall procedure. All four tiers share the same input, label set and evaluation metric; they differ only in how the utterance is represented and in the classifier used on top of it.

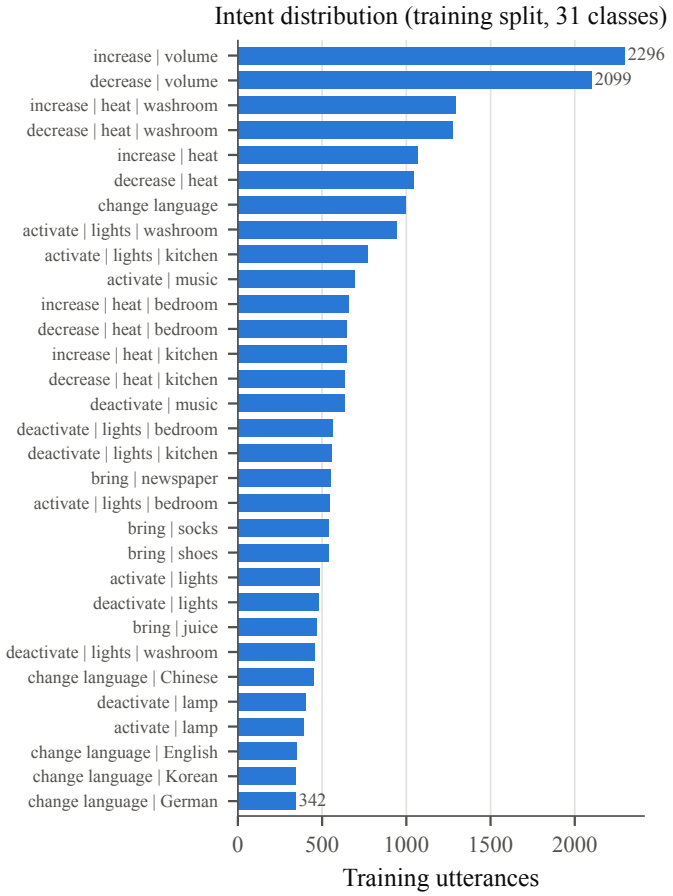


Fig. 1. Number of training utterances per intent (action | object | location; the value *none* is omitted).

A. Pre-processing and short-time analysis

Each signal $x[n]$ is divided into frames of $N = 400$ samples (25 ms) with a hop of 160 samples (10 ms), and each frame is multiplied by the Hamming window

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right), \quad 0 \le n \le N-1. \quad (1)$$

For frame m with samples $x_m[n]$, the short-time energy and zero-crossing rate are

$$E_m = \sum_{n=0}^{N-1} x_m^2[n], \quad (2)$$

$$Z_m = \frac{1}{N-1} \sum_{n=1}^{N-1} \tfrac{1}{2} \left|\operatorname{sgn}(x_m[n]) - \operatorname{sgn}(x_m[n-1])\right|. \quad (3)$$

A frame is labelled as speech when $10 \log_{10} E_m$ is more than 10 dB above the 15th percentile of the frame energies in that utterance. Speech frames whose Z_m is above the 70th percentile of Z over the speech frames are labelled unvoiced, and the rest are labelled voiced. Both thresholds are therefore relative to the recording itself. This is necessary because the absolute signal level varies with the speaker, the microphone and the recording gain across the 97 speakers.

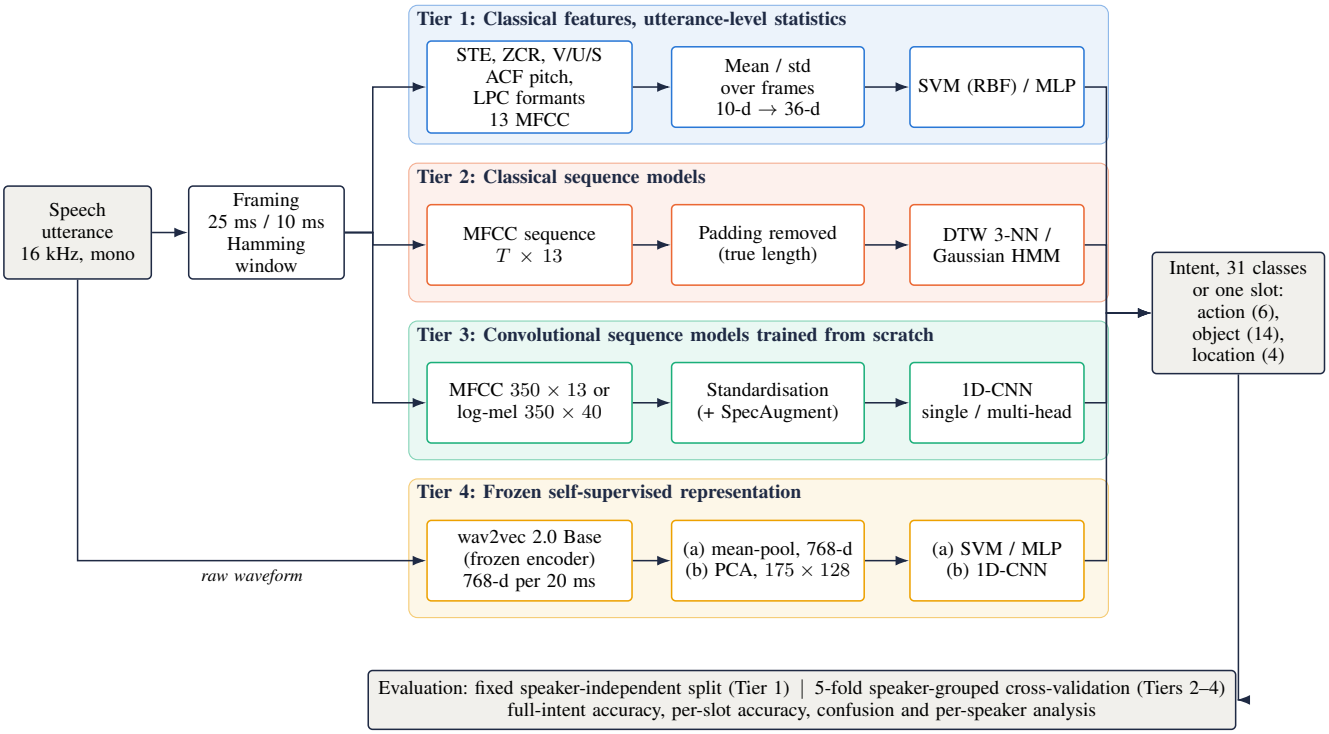


Fig. 2. Methodology. The four tiers are ordered from classical to modern representations; each one produces an intent decision that is evaluated under the protocol shown at the bottom.

For voiced frames, the short-time autocorrelation $r_m[k] = \sum_n x_m[n]\, x_m[n+k]$ is computed and its maximum is searched for over the lags that correspond to 70–400 Hz. The fundamental frequency is then $F_0 = f_s/k^*$, and the ratio $r_m[k^*]/r_m[0]$ is kept as a voicing confidence; estimates below 0.3 are treated as unreliable. Spectrograms are computed as the squared magnitude of the short-time Fourier transform [10]. Linear prediction models each sample as $\hat{x}[n] = -\sum_{k=1}^{p} a_k x[n-k]$. The coefficients of order $p = 16$ are obtained from the autocorrelation of the windowed frame using the Levinson–Durbin recursion [11]. Formant candidates are taken as the peaks of $1/|A(e^{j\omega})|$ below 3.5 kHz with a prominence of at least 0.3 dB, and the prediction residual $e[n] = x[n] - \hat{x}[n]$ gives an approximation of the excitation. MFCCs [12] are computed from the same frames using a mel filter bank, a logarithm and a discrete cosine transform (DCT), and the first 13 coefficients are kept.

B. Tier 1: utterance-level statistics

Each utterance is summarised by ten prosodic statistics. These are the mean and standard deviation of the frame energy (in dB), the zero-crossing rate and F_0; the fraction of voiced frames; the mean first and second formant frequencies over the voiced frames; and the duration. To process the full corpus, F_0 is estimated with the YIN algorithm [9], which is based on the same periodicity principle as the autocorrelation method but is much faster. Adding the mean and standard deviation of each of the 13 MFCCs gives a 36-dimensional vector. All features are standardised using the training-set statistics. An RBF-kernel SVM [27] is tuned over $C \in \{1, 10, 100\}$ and $\gamma \in \{\text{scale}, 0.01, 0.1\}$, and an MLP over hidden layers $\{(64), (128), (64, 32)\}$ and L_2 penalty $\alpha \in \{10^{-4}, 10^{-3}\}$, with the best setting selected on the official validation subset. Inverse-frequency class weighting was also tried. The selected configurations are later evaluated again, without any changes, under the cross-validation protocol of Section V-G.

C. Tier 2: classical sequence models

The MFCC sequences are padded or cut to 350 frames (the 95th percentile of the duration) for storage. However, the padding is removed before sequence modelling, so that each model sees the true length of the utterance. For DTW [14], an approximate multiscale DTW with Euclidean local distance compares each query with five randomly chosen templates per intent (155 in total), and the label is decided by a 3-nearest-neighbour vote. Since each comparison is expensive, DTW is evaluated on 155 queries (five per intent) taken from the held-out speakers of the first cross-validation fold. For the HMM [15], one three-state Gaussian HMM with diagonal covariances is trained per intent using 20 Baum–Welch iterations. A query is assigned to the intent whose model gives the highest log-likelihood.

D. Tier 3: convolutional sequence models

The convolutional network in Fig. 3 works along the time axis and treats the feature dimensions as channels. It is trained from random initialisation with Adam [28] and sparse categorical cross-entropy, using a batch size of 128 and at

most 50 epochs. Training stops early (patience of 8 epochs) based on a separate validation group of speakers. The inputs are standardised per channel using statistics from the training fold. The input representation is changed first, and then the output structure:

(i) a single 31-way head is trained on 13-dimensional MFCC sequences (115,871 parameters);
(ii) the same network is trained on 40-band log-mel sequences computed with the same framing. These differ from MFCC only in that the DCT, and the truncation that follows it, are left out;
(iii) to separate these two operations, the orthonormal DCT-II is applied to the same 40-band log-mel frames, and either all 40 coefficients (an invertible transform) or only the first 13 are kept;
(iv) finally, the output is factorised. In this multi-head variant, a shared trunk on MFCC input feeds three softmax layers for action, object and location. It is trained with the sum of the three losses and with SpecAugment-style masking [21] of two 20-frame time blocks and two 3-coefficient blocks per sequence.

A multi-head prediction is counted as correct only when all three slots are correct.

E. Tier 4: frozen self-supervised representation

The wav2vec 2.0 Base encoder [5] was pre-trained without labels on 960 h of LibriSpeech [29]. It takes the raw waveform as input and produces a 768-dimensional vector every 20 ms from its final transformer layer. Its weights are never updated in these experiments. Two downstream configurations are evaluated. In the pooled configuration, the frame vectors are averaged over time and classified by an RBF-kernel SVM ($C = 10$, $\gamma =$ scale, balanced class weights) and by an MLP with hidden layers (256, 128); these settings are fixed across folds. In the sequence configuration, the frame vectors are projected to 128 dimensions using principal component analysis (PCA). The PCA is fitted on about 300,000 frames from 2,700 utterances of the official training subset and keeps 93.6% of the variance. The projected sequences are padded or cut to 175 frames and classified by the network in Fig. 3. This projection also keeps the cached sequences for the full corpus within the available memory.

F. Reduced-class tasks

Each intent is a combination of three slot values. Hence, three reduced-class tasks follow directly from the label structure: action (6 classes), object (14) and location (4). No intents are removed and no classes are hand-picked; each utterance is simply labelled with one slot value instead of the full intent. The SVM and MLP of Tier 1, the log-mel CNN of Tier 3 and the wav2vec 2.0 sequence CNN of Tier 4 are retrained with a single softmax output for each task. The class distributions are very different across tasks (for example, the value *none* makes up 61% of the locations). Therefore, the majority-class accuracy of each task is reported as its reference.

G. Evaluation protocol

Tier 1 models are reported both on the official partition, so that they can be compared with the corpus paper, and under cross-validation. For cross-validation, the 97 speakers are divided into five folds using grouped K-fold splitting on speaker identity. Thus, no speaker appears on both sides of a fold, and the same speaker folds are used for every model and every task. Each fold holds out about 6,008 utterances from 19–20 speakers. For the networks, 10% of the remaining speakers are set aside for early stopping. The main metric is full-intent accuracy, reported as the mean $\pm$ sample standard deviation over the folds. Since all models share the same folds, they are also compared using fold-wise paired differences and a paired t-test. For the error analysis, per-slot accuracy, confusion matrices and per-speaker accuracy are used.

This protocol was chosen after a preliminary run of the single-head MFCC network on the official partition. This run reached 84.4% on the test subset, while the validation accuracy stayed between 62% and 67%. When the accuracy was examined for each speaker, three of the ten validation speakers were recognised at only 24.1%, 34.9% and 39.8%, whereas every test speaker was above 69%. With only ten speakers per subset, a single-split result depends as much on which speakers were drawn as on the model itself. Hence, speaker-grouped cross-validation over all 97 speakers was used for every model.

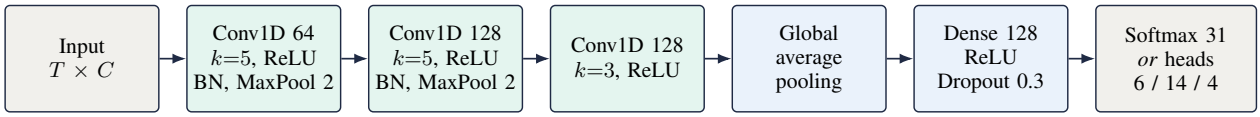
VI. Results and Discussion

A. Short-time analysis of a single utterance

The feature-extraction pipeline was first checked on the utterance *"Bring socks"* (1.76 s). The energy–ZCR labelling gives 75 voiced, 30 unvoiced and 69 silent frames (Fig. 4). The two /s/ fricatives appear as stretches of low energy and high ZCR (0.15–0.21), which is expected from turbulent excitation. The /k/ closure near 1.3 s is partly labelled as silence. This is because the burst is shorter than one analysis frame, so a 25 ms frame mixes the closure, the burst and the voicing. The first attempt used a threshold relative to the maximum energy, and it did not label any frame as unvoiced, since this speaker's fricatives lie close to the noise floor. Setting the threshold relative to the utterance's own noise floor solved this problem.

The autocorrelation pitch analysis (Fig. 4, bottom) gives an F_0 estimate for all 75 voiced frames, and 63 of these are above the confidence threshold. For the frame at 0.752 s, the autocorrelation peaks at a lag of 52 samples ($F_0 = 307.7$ Hz), with further peaks at integer multiples of this lag. The contour rises from about 210 to 310 Hz over *"bring"* and falls from about 258 to 195 Hz over *"socks"*. One estimate at 0.463 s (98.2 Hz, confidence 0.458) is close to half the value of its neighbours. This is an octave error that still passes the confidence test, and it shows that restricting the range and filtering by confidence reduce such errors but do not remove them completely [8].

The spectrogram (Fig. 5) gives an independent check of these measurements. During *"bring"*, the harmonics are



(T, C): MFCC $(350, 13)$; log-mel $(350, 40)$; wav2vec 2.0 PCA $(175, 128)$

Fig. 3. One-dimensional convolutional network used in Tiers 3 and 4. BN denotes batch normalisation.

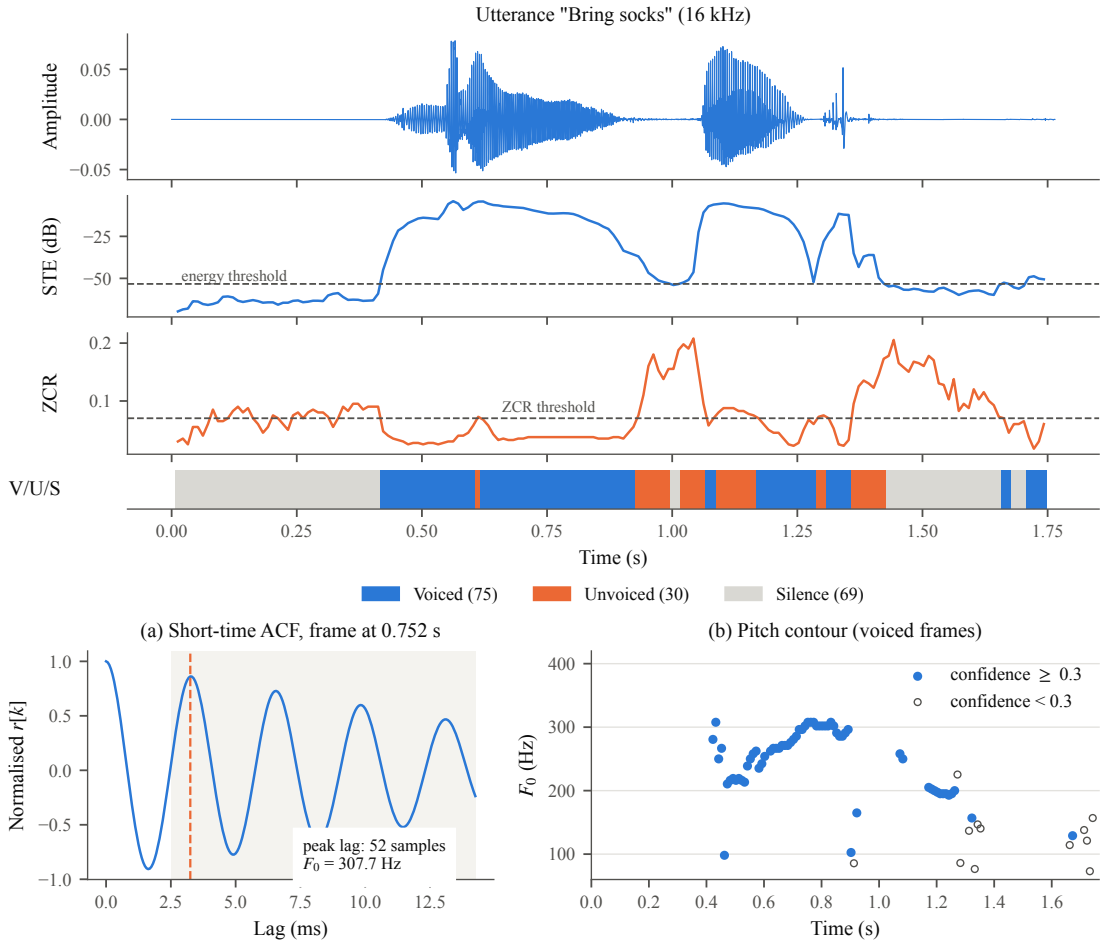


Fig. 4. Short-time analysis of *"Bring socks"*. Top: waveform, short-time energy, zero-crossing rate and the resulting voiced/unvoiced/silence labels. Bottom: (a) normalised autocorrelation of a voiced frame, with the 70–400 Hz search range shaded; (b) frame-wise F_0 contour.

spaced roughly 250–300 Hz apart, which agrees with the autocorrelation estimate. The fricatives show spread-out energy with no harmonic structure, and the /k/ release appears as a short, broadband vertical stripe. The narrowband (40 ms) analysis resolves the individual harmonics, while the wideband (5 ms) analysis shows the individual glottal pulses but loses frequency detail. The order-16 linear-prediction envelope of the frame at 0.752 s (Fig. 5, bottom) resolves two resonances, at 305 and 656 Hz. A separate envelope computed with Burg's method placed both peaks at the same frequencies. The 305 Hz peak is close to the measured F_0 of this frame. It is a plausible first formant for the close front vowel in *"bring"*; however, from a single frame, a contribution from a strong low harmonic at this high F_0 cannot be ruled out. Finally, the normalised autocorrelation of the residual at the pitch lag decreases from 0.406 at $p = 8$ to 0.346 at $p = 20$. This shows that higher prediction orders absorb part of the source periodicity into the all-pole filter.

B. Tier 1: utterance-level statistics

Table III shows the results of the utterance-level classifiers. With prosodic statistics alone, the tuned SVM reaches 16.5% on the test subset, which is 6.1 points above the majority baseline. For both classifiers, the accuracy levelled off at around 15–16% across the hyperparameter grid. This suggests that the limitation lies in the features rather than in the classifier. The confusion matrix (Fig. 6) shows two clear patterns. First, the most frequent intents attract many of the

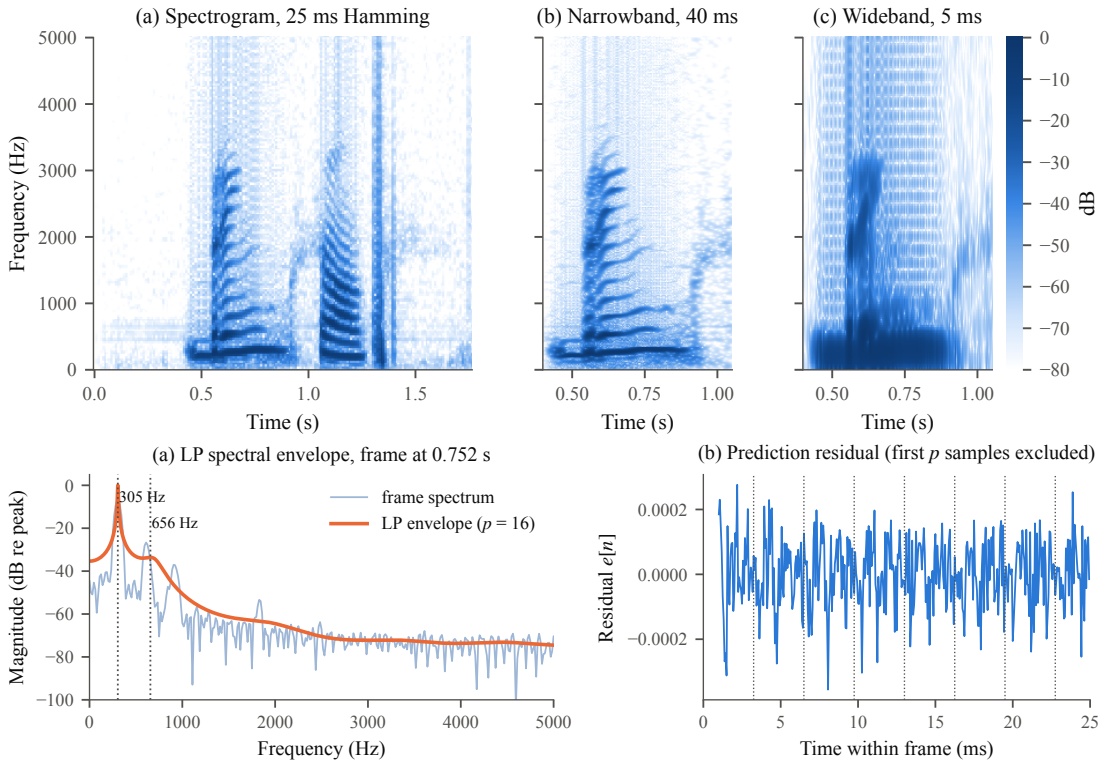


Fig. 5. Spectral analysis of *"Bring socks"*. Top: (a) 25 ms Hamming spectrogram of the full utterance; (b) narrowband and (c) wideband analysis of *"bring"*. Bottom: (a) order-16 linear-prediction envelope over the spectrum of the frame at 0.752 s; (b) prediction residual, with dotted lines at multiples of the measured pitch period.

TABLE III
UTTERANCE-LEVEL CLASSIFIERS, 31-CLASS ACCURACY (%): OFFICIAL PARTITION AND 5-FOLD SPEAKER-GROUPED CV

Features	Classifier	Valid.	Test	CV
–	Majority class	10.2	10.4	10.0 ± 0.2
Prosodic	SVM (C=10, γ=0.01)	16.4	16.5	17.8 ± 1.0
Prosodic	MLP (64)	15.2	14.3	16.5 ± 1.5
+ MFCC	SVM (C=1, γ=0.01)	22.3	**25.8**	$\mathbf{24.0 \pm 1.0}$
+ MFCC	MLP (64, 32)	22.4	21.0	21.4 ± 1.0

predictions: *increase volume* makes up 20.3% of the SVM's predictions but only 10.4% of the test labels. Second, the best-recognised intent, *change language Chinese* (52.0% recall), is identified mainly by its duration, since its phrasings last 3.06 s on average compared with a training mean of 2.29 s. Eight intents are recognised at or below 1%. Inverse-frequency class weighting increased the SVM's test macro-F1 only from 0.115 to 0.120 and reduced its accuracy to 12.2%, so it was not used.

Adding the MFCC statistics raises the SVM to 25.8% (+9.3 points) and the MLP to 21.0% (+6.6 points). Under speaker-grouped cross-validation, the same SVM configurations give $17.8 \pm 1.0\%$ and $24.0 \pm 1.0\%$, and the MFCC statistics improve every fold (mean paired difference 6.2 points, $p < 0.001$). Thus, the conclusions drawn from the official partition also hold under the stricter protocol. This gain is larger than any

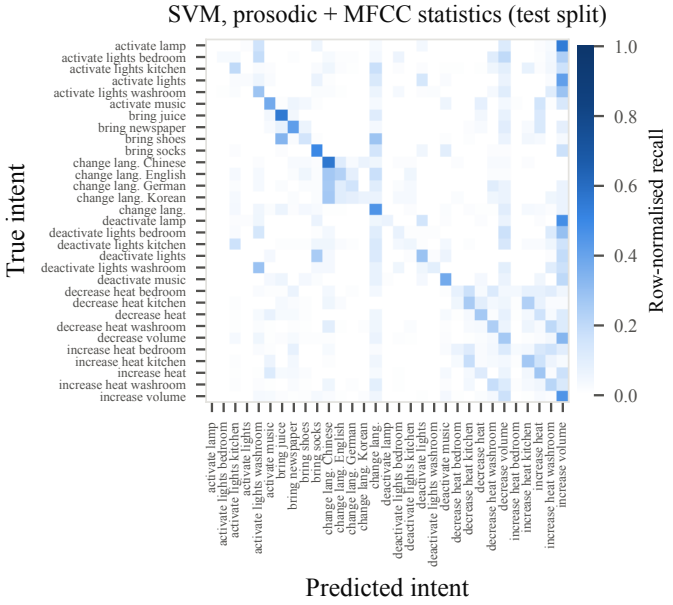


Fig. 6. Row-normalised confusion matrix of the SVM on prosodic and MFCC statistics (official test subset, 25.8% accuracy).

improvement obtained by tuning the prosodic system, and it fits the difference between the two kinds of features. Energy, ZCR, pitch, the two formants and the duration describe *how* an utterance is spoken. The cepstral envelope, on the other

TABLE IV
SPEAKER-GROUPED FIVE-FOLD CROSS-VALIDATION (FULL-INTENT ACCURACY, %)

Model	Mean ± s.d.	Range
Prosodic + MFCC statistics + SVM (Tier 1)	24.0 ± 1.0	22.7–25.2
Gaussian HMM, MFCC	6.0 ± 0.7	5.3–7.1
1D-CNN, MFCC	72.5 ± 5.0	65.4–77.6
1D-CNN, log-mel	79.4 ± 6.8	70.9–88.5
Multi-head 1D-CNN, MFCC	77.3 ± 6.7	67.6–83.3
wav2vec 2.0 pooled + SVM	59.9 ± 4.4	55.8–67.5
wav2vec 2.0 pooled + MLP	69.9 ± 4.7	66.3–78.0
wav2vec 2.0 sequence + 1D-CNN	$\mathbf{93.1 \pm 3.1}$	88.9–96.9

TABLE V
EFFECT OF THE CEPSTRAL DCT ON THE 1D-CNN (31-CLASS ACCURACY, %, 5-FOLD SPEAKER-GROUPED CV)

Input (per frame)	Dim.	Mean ± s.d.
40-band log-mel	40	$\mathbf{79.4 \pm 6.8}$
DCT of log-mel, first 13 coefficients	13	73.1 ± 5.8
MFCC (128 mel bands), 13 coefficients	13	72.5 ± 5.0
DCT of log-mel, all 40 coefficients	40	63.5 ± 5.0

hand, carries information about *which* phones occur, and it is the phones that decide the intent. However, averaging over the utterance still removes the order in which the phones occur, and this limits what any static feature vector can achieve.

C. Tier 2: classical sequence models

The DTW 3-NN classifier reaches 11.6% on its 155-query subset, and the per-class Gaussian HMM reaches $6.0 \pm 0.7\%$ over the five folds. Both are close to or below the majority baseline. When the HMMs were trained and scored on the zero-padded sequences, the accuracy was 9.8%. The drop after removing the padding shows that the padded version was partly using the amount of padding, which is a duration cue related to the intent, rather than the spectral structure. When the scores were inspected after trimming, the log-likelihoods were finite and well separated, but a few class models won a disproportionate share of the queries. These baselines were deliberately kept simple (three ergodic states, one Gaussian per state, no dynamic features and five DTW templates per class). Hence, the comparison shows that such configurations are far from enough for 31 multi-word intents with 248 phrasings; it does not show that classical sequence modelling is unsuitable for the task in general.

D. Tier 3: convolutional sequence models

Table IV summarises the cross-validated models, grouped by tier. The single-head CNN on MFCC sequences reaches $72.5 \pm 5.0\%$, which is three times the static MFCC result of Tier 1 on the same folds. Clearly, temporal order was the main missing factor in Tier 1.

When the MFCC input is replaced with 40-band log-mel input in the same single-head network, the accuracy rises to $79.4 \pm 6.8\%$, with an improvement in every fold (mean 6.9 points, $p = 0.005$). The two representations share the mel filter bank and the logarithm. MFCC adds a DCT and a truncation to 13 coefficients, and the log-mel input is also wider (40 against 13 dimensions). Table V separates these factors by applying the DCT to the same log-mel frames. Keeping all 40 DCT coefficients, which preserves every dimension and is exactly invertible, lowers the accuracy to $63.5 \pm 5.0\%$. This is 15.9 points below log-mel in every fold ($p < 0.001$). Truncating the same transform to 13 coefficients brings the accuracy back to $73.1 \pm 5.8\%$, which cannot be distinguished from the original MFCC computed from 128 mel bands ($p = 0.60$). Therefore, the advantage of the log-mel input comes from leaving out the DCT and not from its extra dimensions, and the size of the mel filter bank has no measurable effect.

In principle, the first convolutional layer could learn to undo an invertible linear map. Hence, the loss must come from the way the transformed input is presented to the network. Before training, each input channel is standardised to unit variance. Across the log-mel bands, the standard deviations differ by a factor of 1.9; across the DCT coefficients, they differ by a factor of 60, and coefficients 14–40 carry only 2.2% of the total variance. Standardisation therefore gives these low-variance, high-quefrency coefficients, which mostly describe fine spectral detail and noise, the same weight as the coefficients that describe the spectral envelope. Truncation removes most of them, and this is why 13 coefficients do better than 40. The DCT is still suitable for diagonal-covariance Gaussian models, which benefit from decorrelated inputs. For the convolutional classifier used here, however, the log-mel representation is the better choice, which agrees with the findings of Mohamed *et al.* [17].

Once the input was fixed, the output structure was changed. Splitting the MFCC network into action, object and location heads, and training it with SpecAugment masking, raises the accuracy to $77.3 \pm 6.7\%$. This is an improvement over the single-head MFCC network in every fold (mean paired difference 4.8 points, $p = 0.007$). The per-slot accuracies are 85.3% for the action, 91.7% for the object and 94.1% for the location (Fig. 7), so the action slot is the weakest. Since this variant also adds augmentation, the gain comes from the combination of the factorised output and SpecAugment. The single-head log-mel network and the multi-head MFCC network do not differ consistently (2.1 points in favour of log-mel, in 4 of 5 folds, $p = 0.16$). Hence, the two changes give a similar benefit; their combination was not tested.

E. Tier 4: frozen wav2vec 2.0 representation

With mean pooling, the frozen wav2vec 2.0 representation gives $59.9 \pm 4.4\%$ with the SVM and $69.9 \pm 4.7\%$ with the MLP. These results are much higher than the static MFCC statistics of Tier 1. However, the pooled MLP is not significantly different from the MFCC CNN trained from scratch (mean paired difference -2.6 points, $p = 0.24$), since pooling throws away the temporal order just like the Tier 1 statistics do. When the frame sequence is kept and classified by the network

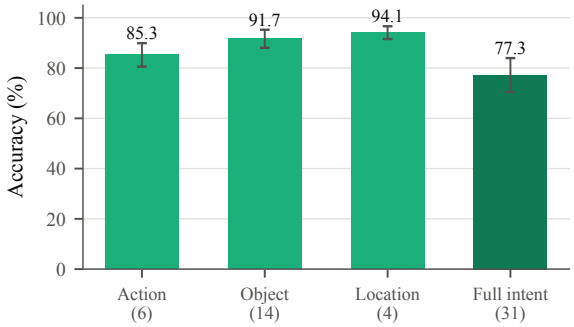


Fig. 7. Per-slot and full-intent accuracy of the multi-head CNN (mean ± s.d. over folds; number of classes in parentheses).

in Fig. 3, the accuracy rises to $93.1 \pm 3.1\%$. This is higher than the log-mel CNN in every fold (mean paired difference 13.7 points, $p = 0.001$), and it also has the smallest spread across folds of all the models. Thus, the pooled and sequence results separate two effects: the pre-trained representation is the stronger feature when the temporal model is kept the same, but temporal modelling is still needed to make use of it.

The cross-validated accuracy of the frozen encoder is close to the 92.35% that SUPERB reports for frozen wav2vec 2.0 Base on the official test split [4], although the evaluation protocols are different. It is still below the 98.8% of the ASR-pre-trained model of Lugosch *et al.* [2] and the 98.34% of frozen HuBERT Base [4]. The present setup uses only the final transformer layer, which is known to carry less phonetic information than the intermediate layers [26], along with a PCA projection that discards 6.4% of the variance. Both of these are likely reasons for the gap.

F. Overall comparison

Fig. 8 places all the 31-class configurations on one axis, ordered from classical to modern. Three steps explain most of the progression: adding cepstral information to the utterance statistics (17.8 to 24.0%), keeping the temporal order with a convolutional network (24.0 to 72.5–79.4%), and replacing the hand-crafted frame features with a frozen self-supervised representation (79.4 to 93.1%). All results except DTW use the same speaker folds; the DTW result is based on a 155-query subset of one fold. In addition, the PCA used in Tier 4 was fitted on utterances from the official training speakers, and some of them fall in held-out folds. Since the projection is unsupervised and generic, this effect is expected to be small, but the procedure is not strictly fold-internal.

G. Reduced-class slot tasks

Table VI and Fig. 9 report the three slot tasks of Section V-F together with the 31-class task, using the same speaker folds. Reducing the number of classes increases the accuracy of every model. For the SVM on prosodic and MFCC statistics, the accuracy goes up from 24.0% on the intent to 40.7% on the action, 49.3% on the object and 64.3% on the location. However, these gains mostly follow the rise of the majority-class baseline, which goes from 10.0% to 25.8%, 31.2% and 61.2%. To compare tasks with different baselines, the fraction of the majority-class error that a model removes is measured, $(a - a_{\mathrm{maj}})/(100 - a_{\mathrm{maj}})$. By this measure, the SVM removes 15.6% of the error on the intent, 20.1% on the action, 26.3% on the object and only 8.0% on the location. On the location task, the MLP even falls below the baseline (60.5% against 61.2%). In most utterances the location is not spoken at all, and when it is, it is a single word near the end of the command; this is a cue that averages over the utterance capture poorly.

TABLE VI
REDUCED-CLASS SLOT TASKS: MEAN ACCURACY (%) OVER FIVE SPEAKER-GROUPED FOLDS (NUMBER OF CLASSES IN PARENTHESES)

Model	Loc. (4)	Act. (6)	Obj. (14)	Intent (31)
Majority class	61.2	25.8	31.2	10.0
Prosodic + SVM	62.6	35.8	42.9	17.8
Prosodic + MFCC + SVM	64.3	40.7	49.3	24.0
Prosodic + MFCC + MLP	60.5	34.3	42.4	21.4
1D-CNN, log-mel	90.6	85.1	89.1	79.4
wav2vec 2.0 + 1D-CNN	**97.5**	**95.1**	**96.4**	**93.1**

The sequence models behave quite differently. The log-mel CNN reaches 85.1–90.6% on the slot tasks and the wav2vec 2.0 network reaches 95.1–97.5%. They remove 76–84% and 92–95% of the majority-class error respectively, with little difference between tasks. Therefore, a smaller label set raises the headline accuracy of the classical pipeline, but it does not change its position relative to the other tiers. The limiting factor is the loss of temporal information in the utterance-level representation, not the number of classes. It is also observed that the single-slot log-mel networks do not do better than the matching heads of the multi-head MFCC network (85.1, 89.1 and 90.6% against 85.3, 91.7 and 94.1% for action, object and location). This suggests that training the slots jointly, along with augmentation, helps each individual slot.

H. Error analysis of the best model

To look at individual errors, the wav2vec 2.0 sequence model was retrained on the first fold and its predictions were saved (91.7% accuracy, compared with 91.4% in the original run). The confusion matrix (Fig. 10) is strongly diagonal, and most of the remaining errors are between commands of opposite polarity. *Increase volume* is predicted as *decrease volume* 40 times, *increase heat* is predicted as *decrease heat* in every location (8–14 times per location), and similar confusions occur for the music and lamp commands. Accordingly, the slot accuracy is lowest for the action (93.9%), compared with 95.2% for the object and 97.2% for the location. In these phrasings, the polarity is often carried by one short, unstressed word (*up*/*down*, *on*/*off*) while the rest of the utterance is the same, so a small acoustic cue decides the label. The accuracy also varies widely across the 19 held-out speakers: 14 of them are above 89%, while two are recognised at only 36.4% and 44.2%. This kind of speaker variability explains both the spread across folds in Table IV and the instability of the ten-speaker official split.

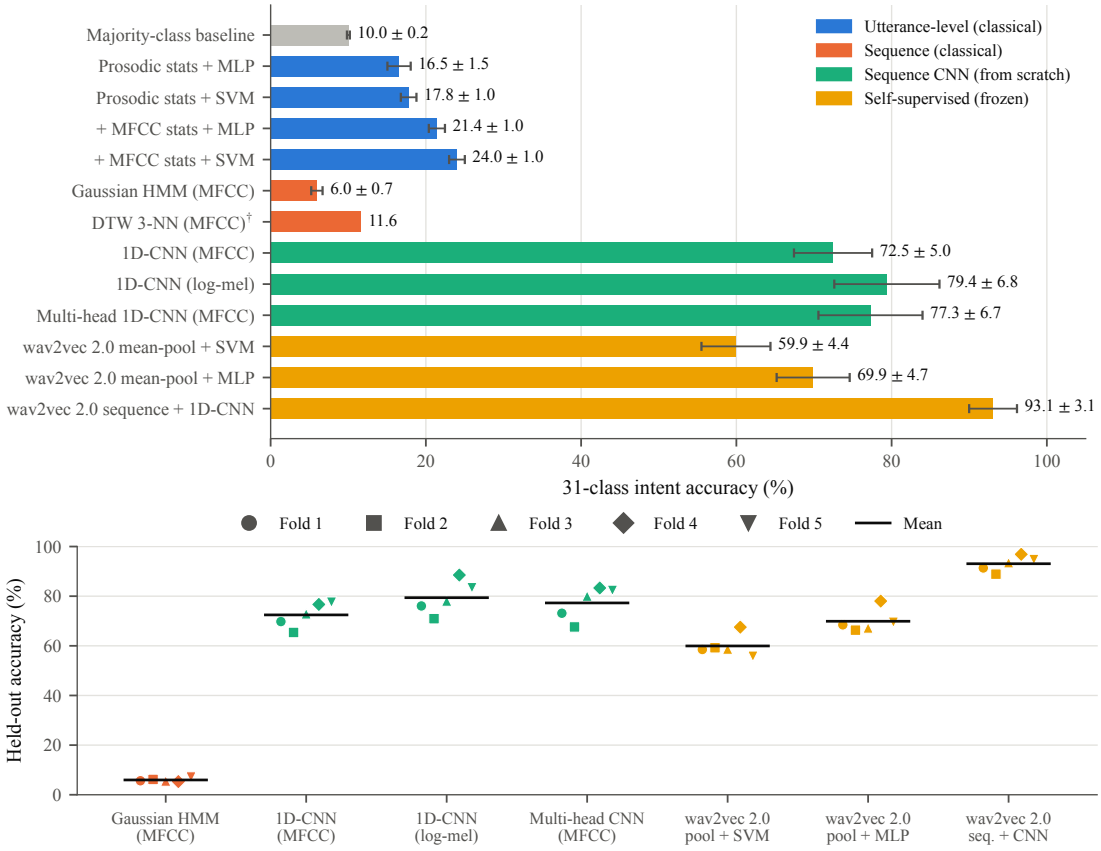


Fig. 8. Top: intent accuracy of all configurations under 5-fold speaker-grouped cross-validation, ordered from classical to modern; error bars show the standard deviation over folds (†DTW is evaluated on a 155-query subset of one fold). Bottom: held-out accuracy of each model per fold, with horizontal bars at the fold mean.

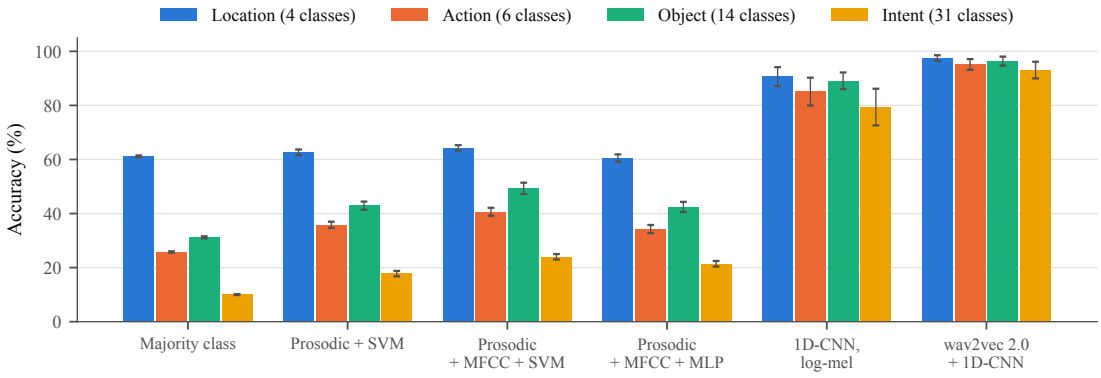


Fig. 9. Accuracy on the three reduced-class slot tasks and on the 31-class intent task (mean ± s.d. over five speaker-grouped folds).

VII. Conclusion

In this study, spoken intent classification on Fluent Speech Commands was examined using a sequence of representations, from classical short-time analysis to a frozen self-supervised encoder. The classical feature-extraction pipeline behaved as expected on individual utterances, and its pitch, spectral and linear-prediction measurements agreed with one another. However, prosodic utterance statistics carry little of the information that separates the intents (17.8% under cross-validation), and cepstral statistics recover only part of it (24.0%). Simple DTW and HMM baselines did not do better than the majority class. Convolutional networks trained from scratch on frame sequences raise the accuracy to 72.5–79.4%. Among these, log-mel input does better than MFCC in every fold, because the cepstral DCT combined with per-channel standardisation inflates the low-variance coefficients. The same network on frozen wav2vec 2.0 frames reaches 93.1%.

Across the whole progression, keeping the temporal order

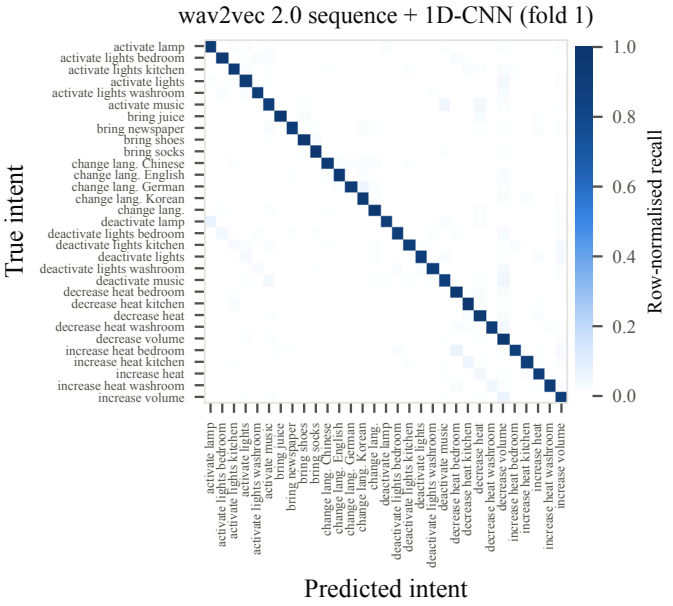


Fig. 10. Row-normalised confusion matrix of the wav2vec 2.0 sequence model on the held-out speakers of fold 1 (91.7% accuracy).

gave the largest single gain, and the pre-trained representation gave the second largest. Reducing the task to the action, object or location slot increased the accuracy of every model. However, the utterance-level statistics removed at most a quarter of the majority-class error on any slot, while the sequence models removed 76–95%. Hence, the number of classes is not the main difficulty for the classical pipeline. The remaining errors of the best model are mostly between commands of opposite polarity and on a few difficult speakers. The evaluation also showed that the ten-speaker official subsets of FSC are too small for stable single-split results, and that speaker-grouped cross-validation with paired comparisons is a more reliable way to compare models on this corpus.

There are several ways to extend this work. Classical sequence baselines with a left-to-right topology, Gaussian mixtures and dynamic features [13] would give a fairer classical reference. For the self-supervised tier, the natural next steps are a learned weighting of the encoder layers [4], a comparison with HuBERT [25] and fine-tuning of the encoder, along with evaluation under additive noise and on unseen phrasings.

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